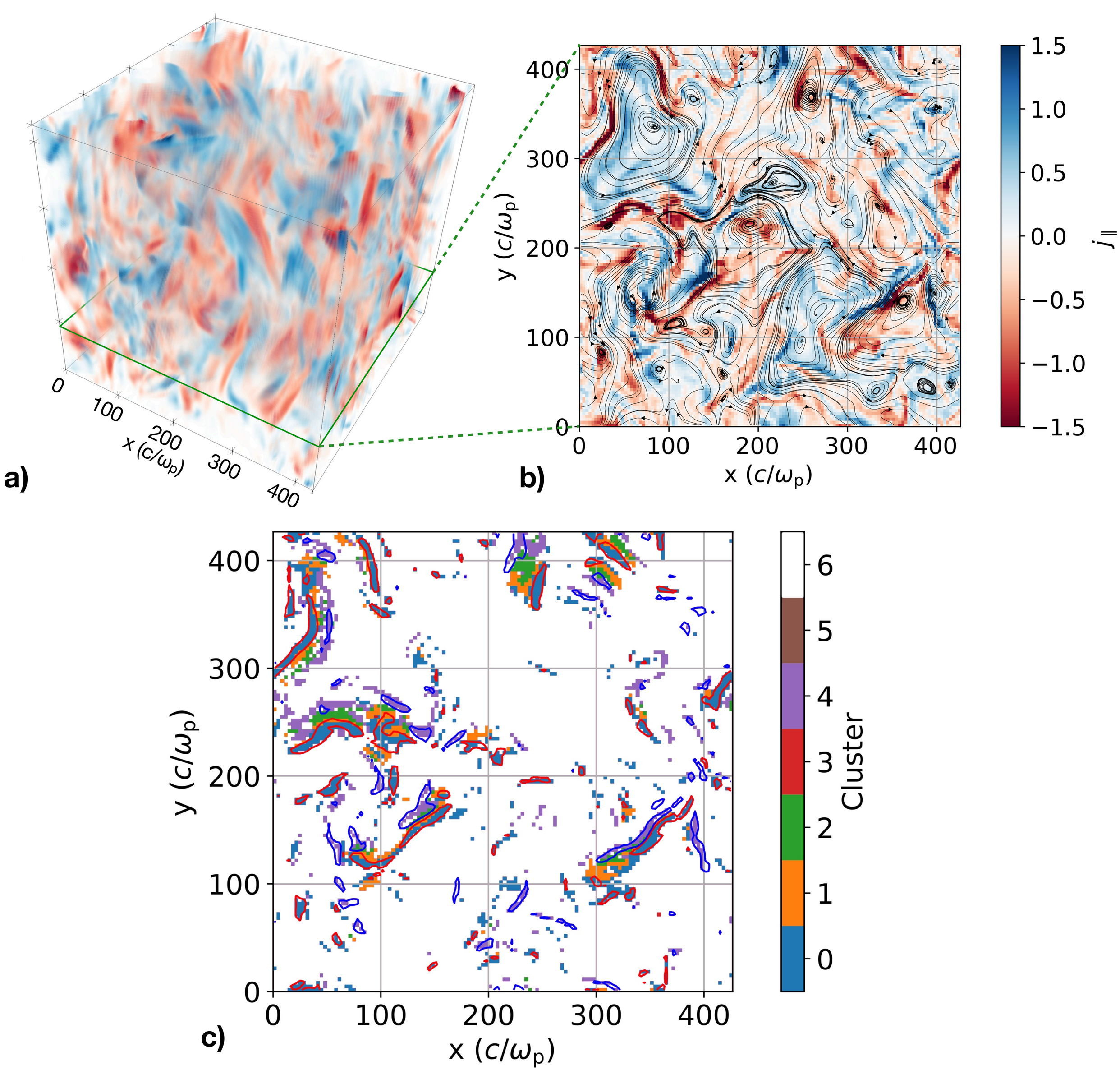

a)
0
100
200
300
400
x (c/ω_p)
b)
y (c/ω_p)
x (c/ω_p)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
j_∥
c)
y (c/ω_p)
x (c/ω_p)
Cluster
0
1
2
3
4
5
6